DUSHYANT MAHAJAN

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Education

Northeastern University **Sep 2022 – Dec 2024**
Master of Science, Computer Software Engineering | ***GPA**: 3.67/4.00* *Boston, MA*

University of Mumbai **Aug 2015 – Oct 2020**
Bachelor of Technology, Computer Science and Engineering *Mumbai, India*

Professional Experience

Raga AI, Fremont, CA **Jan 2024 - Jul 2024**
Data Scientist Co-op *[HuggingFace, Langchain, LlamaIndex, Crew AI, DSPy, Pydantic]*

- Engineered a **LLM evaluation** framework to rigorously assess model quality across 50+ evaluation **metrics and guardrails**, establishing crucial quality benchmarks for production RAG applications and improving LLMs response accuracy
- Collaborated to **open-source** Raga AI Catalyst, an **observability** platform that aids **RAG system** debugging by providing granular insights through **timeline and execution graphs**, facilitating to identify and resolve performance bottlenecks
- Architected RAG Builder, a modular package that streamlined RAG development by unifying document loaders, **LLM APIs**, and vector databases, dramatically reducing the time required for prototyping custom RAG pipelines
- Led end-to-end testing of RAG and **Agentic workflows** across multiple LLM providers (OpenAI, Vertex AI, Ollama, Groq), developing optimal prompting strategies and establishing performance benchmarks for production deployments

Data Science Consultant *[Docker, Fast API, Tensorboard, Google Cloud, PyTorch, Plotly]* **May 2022 – Aug 2022**

- Designed an **interactive analytics** dashboard using Fast API that transformed image embedding data into clusters and actionable insights, enabling stakeholders to visualize and interpret high-dimensional datasets
- Developed a sophisticated **data drift detection** system to uncover four critical model biases like selection bias, in classification image datasets, implementing MMD and Kolmogorov-Smirnov methods to ensure **model reliability** and performance stability
- Created Docker images and testing suites for **CI/CD** pipelines with ML Flow and DVC (Data Version Control) integration
- Transformed client's image processing workflow by implementing **CLIP-based semantic search**, reducing image tagging processing time by 70% while maintaining high accuracy and relevance

Askim Technologies, Mumbai, India **May 2020 – May 2022**
Founding Software Engineer *[ReactJS, NodeJS, MongoDB, AWS, Python, Golang, Packer, Pulumi, SQL]*

- Led **DevOps** initiatives by architecting and deploying a highly available MERN stack application on AWS, implementing multi-AZ infrastructure that achieved 99.9% uptime
- Implemented secure **authentication** and access control protocols using **JWT** and with HTTPS-enabled **CRUD** endpoints
- Integrated Docker containers with AWS **Lambda** functions for **serverless** processing of image and video generation pipeline
- Increased click-through rate by 15% through **A/B testing** UI/UX features utilizing **CloudWatch** logging metrics for monitoring
- Streamlined an automated AMI generation pipeline using HashiCorp Packer and **GitHub Actions**, establishing a reliable CI/CD process that reduced deployment errors and infrastructure updates
- Employed **infrastructure as code** (IaC) using Pulumi to automate the provisioning of comprehensive AWS environments, including **VPC, EC2, RDS, Lambda**, and monitoring services, enabling consistent and reliable infrastructure deployment

Projects

Reducing EHR (Electronic Health Record) Chart Burden using LLMs | ACL Paper Link

- Researched and improved a LLM pipeline for summarizing **Electronic Health Records** (EHRs) from MIMIC-IV dataset, reducing hallucinations, and improving chunk retrieval accuracy utilizing prompt engineering and context-length optimization
- Engineered RAG workflows with robust QA framework, attaining **86% F1 BERT Score** using locally hosted **Llama models**, reducing clinician chart review times by 30% and maintaining **patient data privacy**

Cloud Automation for Golang App *[Golang, GORM, Pulumi, AWS]*

- Developed user and product management **Golang APIs** with Basic Auth authentication, secure password storage using **BCrypt** encryption, and **GORM** for database operations
- Leveraged Pulumi **IaC** to establish to establish networking infrastructure on AWS compute, RDS and storage, for efficient provisioning and resource management

Fine-Tuning Stable Diffusion for Synthetic Skin Rashes Image Generation *[Streamlit, Diffusers, Stable Diffusion, PyTorch]*

- Created a **Streamlit** application capable of generating high-fidelity images reflecting diverse rash characteristics across 50+ variations in skin tone and anatomical locations, improving educational tools for dermatology professionals
- Fine-tuned **Stable Diffusion** using Dreambooth to generate text embeddings, conditioned the U-Net on those embeddings to generate four realistic and detailed image variations per input query minimizing **FID score**